3.2	(38) $\cos\alpha' = \dfrac{\sqrt{1-V^2/c^2}\cos\alpha}{1-V\sin\alpha/c} \approx \cos\alpha\left(1+V\sin\alpha/c\right)$ $\sin\alpha' = \dfrac{\sin\alpha - V/c}{1-V\sin\alpha/c} \approx \sin\alpha - V\cos\alpha^2/c$	0,2	**1,4**
	(39) $\sin\alpha' = n\sin\beta'$	0,2	
	(40) $v_{ph} = \dfrac{c}{n}$	0,2	
	(41) $v_m \sin\beta = \dfrac{v_{ph}\sin\beta' + V}{1+\dfrac{v_{ph}V\sin\beta'}{c^2}} \approx v_{ph}\sin\beta' + V$	0,2	
	(42) $v_m \cos\beta = \dfrac{\sqrt{1-V^2/c^2}}{1+\dfrac{v_{ph}V\sin\beta'}{c^2}} v_{ph}\cos\beta' \approx v_{ph}\cos\beta'$	0,2	
	(44) $A_1 = \dfrac{1}{n}\sin\alpha$	0,2	
	(45) $B_1 = -\dfrac{n^2+\cos 2\alpha}{n}$	0,2	
3.3	(47) $A_2 = \dfrac{c}{n}$	0,2	**0,4**
	(48) $B_2 = \left(1-\dfrac{1}{n^2}\right)\sin\beta$	0,2	
3.4	(49) $v_+ = \dfrac{c}{n} + V\left(1-\dfrac{1}{n^2}\right)$	0,2	**0,9**
	(50) $v_- = \dfrac{c}{n} - V\left(1-\dfrac{1}{n^2}\right)$	0,2	
	(51) $\Delta t = \dfrac{2L}{v_-} - \dfrac{2L}{v_+} \approx \dfrac{4Lv\left(n^2-1\right)}{c^2}$	0,2	
	(52) $\Delta l = c\Delta t = \dfrac{4Lv\left(n^2-1\right)}{c}$	0,2	
	(53) $\Delta N = \dfrac{\Delta l}{\lambda} = \dfrac{4Lv\left(n^2-1\right)}{c\lambda}$	0,1	
3.5	(54) $n = \sqrt{1+\dfrac{c\lambda\Delta N}{4Lv}}$	0,1	**0,2**
	$n = \sqrt{1+\dfrac{c\lambda\Delta N}{4Lv}} = 1.37$	0,1	
합계			**10,0**